

Module Schedule

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The schedule below maps each module to its slides, activity loop, weekly **discussion**, and assessment. All seven modules' slide decks are published; dates are week-paced (this term runs 7.5 weeks with no fixed calendar dates).

Each **checkpoint quiz** is due the week its module is taught and **gates the project milestone due the following week**, so the derivation always lands before the milestone that uses it. Week 5 has no discussion: it is the project midpoint.

Week	Topic	Slides	Activity Loop	Discussion	Readings	Assessment
1	Linear Models & Least Squares	Module 1	week01-least-squares	Introductions	PML Ch. 1 (§1.1–1.3) & §1.5 (data); MML Ch. 2 & Ch. 9 (§9.1–9.2)	Prerequisite quiz; HW Units 0–1; The project assigned
2	Generalization	Module 2	week02-bias-variance	The tradeoff in your own words	PML §4.7 (bias-variance) & §11.3 (ridge); MML Ch. 5 (review)	HW Units 2a–2b; Checkpoint quiz 1.1 — normal equations; project teams formed
3	MLE & Parameter Uncertainty	Module 3	—	What likelihood buys us	PML Ch. 2–3; §4.1–4.2 (MLE, incl. §4.2.5) & §4.7 (Fisher info); MML Ch. 6 (review)	HW Unit 3; Checkpoint quiz 1.2 — MLE; Milestone 1.1 due
4	Bayesian Inference	Module 4	week03-bayesian-updating	How the posterior contracts	PML §4.6 (Bayes & conjugacy) & §11.7 (Bayesian linear regression); Think Bayes	HW Unit 4; Checkpoint quiz 1.3 — posterior; Milestone 1.2 due

Week	Topic	Slides	Activity Loop	Discussion	Readings	Assessment
					Ch. 2 & Ch. 4 (supp.: Ch. 18)	
5	Ap-prox-imate Inference & Bayesian Clas-sifica-tion	Module 5	week04-logistic-regression	— (tool only; project mid-point)	PML §10.1–10.2 (logistic; review Ch. 9) & §4.6.8 (Laplace); Think Bayes Ch. 19 (MCMC); optional: MML Ch. 7 (Newton/IRLS)	HW Unit 5; Checkpoint quiz 2.1 — Newton–Raphson; Milestones 1.3 & 1.4 due; Project midpoint: Part 1 due ; Part 2 begins
6	SVM & Clus-tering	Module 6	week05-kmeans	When clusters match labels	PML §21.3–21.4 (k-means, GMM/EM); MML Ch. 11 — SVM lecture-only (optional pointer: PML §17.3)	Checkpoint quiz 2.3 — k-means/PCA; Milestones 2.1 & 2.2 due (Gate A)
7	PCA	Module 7	week06-pca	The course arc	MML §4.2 (eigende-composition) & Ch. 10 (PCA); PML §20.1 (secondary)	Milestones 2.3 & 2.4 due (Gate B)
7.5	Course close	—	—	—	—	The project due: milestone 2.5 and the Part 2 report

(Activity slugs are week-named and land one week after their module opens from Module 4 on — modules ≠ weeks. Module 3 has no interactive tool — its *parameter-uncertainty demo* is used in lecture, not as a loop, so that week’s *discussion* carries the loop on its own. Cross-validation (M2) is lecture-only — no assigned reading.)

Where next (optional — not a module, no assessment): Neural networks (UDL Ch. 2–3, Ch. 6–7) continue in INFO 557.

Legend

- **Slides:** Links to module landing pages listing the live reveal.js decks (dark/light toggle, top-right; a rendered .pdf ships alongside each deck).
- **Activity Loop:** Interactive tool + retakeable mastery quiz → reflection → peer review → revise (completion tokens honor-verified).
- **Readings:** all open-licensed — **PML** = Murphy, *Probabilistic Machine Learning: An Introduction* (2022); **MML** = Deisenroth, Faisal & Ong, *Mathematics for Machine Learning*; **Think Bayes** = Downey, *Think Bayes* (2e); **UDL** = Prince, *Understanding Deep Learning*. PML/UDL excerpt links are hosted with this site; MML, Think Bayes, and un-excerpted PML sections (§11.3, §11.7, §17.3) link to the books' free official pages. Section numbers follow PML1 (2022) — verify against your printing.
- **Assessment:** Homework, checkpoint quizzes (each gating the next week's milestone), milestone due weeks, and project deadlines. Week pacing; no fixed calendar dates.

All external links open in a new tab.